

Table 2.2 also tells us whether the representation is real, complex or quaternionic.

For the spinor representations, since Spin is a subspace of the even Clifford algebra, we only need to look at table 2.2, which implies that the number of irreducible inequivalent spinor representations is

$$s_{r,s} = \begin{cases} 2 & \text{if } r - s = 2, 4 \bmod 4, \\ 1 & \text{if } r - s = 1, 3 \bmod 4. \end{cases}$$

Notice that in the even dimensional case $D = 2k$ there are two inequivalent irreducible spinor representations (known as Weyl representations), which correspond to the Weyl spinors and can be understood by looking at the volume element $v_* = v_1 \cdots v_D$. Being v_* the product of an even number of generators, it anticommutes with them, i.e. $\{v_*, v_a\} = 0$ for all $a = 1, \dots, D$, but it commutes with all the elements in the even Clifford subalgebra, hence with all the group elements in $\text{Spin}(V)$, which implies that it must act as a scalar in the spinor representations. This means that the inequivalent Weyl representations can be labelled by the eigenvalues of v_* . Furthermore, a straightforward computation gives, for even $D = r + s = 2k$

$$v_*^2 = (-1)^{\frac{r-s}{2}} \mathbb{1}. \quad (15)$$

One can classify the inequivalent spinor representation as follows

- $r - s = 0 \bmod 8$. There are two inequivalent real spinor representations, of real dimension $2^{\frac{D-2}{2}}$, labeled by the eigenvalue of v_* being 1 or -1 ;
- $r - s = 1, 7 \bmod 8$. There is a unique spinor representation, which is real and of real dimension $2^{\frac{D-1}{2}}$;
- $r - s = 2, 6 \bmod 8$. There are two inequivalent complex spinor representations, of complex dimension $2^{\frac{D-2}{2}}$, labeled by the eigenvalue of v_* being i or $-i$;
- $r - s = 3, 5 \bmod 8$. There is a unique spinor representation, which is quaternionic and of quaternionic dimension $2^{\frac{D-3}{2}}$;
- $r - s = 4 \bmod 8$. There are two inequivalent quaternionic spinor representations, of quaternionic dimension $2^{\frac{D-4}{2}}$, labeled by the eigenvalue of v_* being 1 or -1 ;

2.4.1 Complex representations and the Lorentzian signature case

As it is significantly easier to deal with complex Clifford algebras, we turn our attention to complex representations of $\mathcal{C}(D)$.

We recall

$$\mathcal{C}(2k) \simeq \mathbb{C}(2^k) \quad \text{and} \quad \mathcal{C}(2k+1) \simeq \mathbb{C}(2^k) \oplus \mathbb{C}(2^k), \quad (16)$$

which implies there are faithful representations

$$\Gamma_{(2k)} : \mathcal{C}(2k) \rightarrow \text{End}(\mathbb{C}^{2^k}) \quad (17)$$

$$\Gamma_{(2k+1)} : \mathcal{C}(2k+1) \rightarrow \text{End}(\mathbb{C}^{2^k}) \oplus \text{End}(\mathbb{C}^{2^k}), \quad (18)$$

where Γ_{2k} is irreducible and Γ_{2k+1} splits into two irreducible representations. These are precisely the pinor representations of the complex Clifford algebra.

Remark 18. The above irreducible representations are unique up to conjugacy with unitary matrices. From now on, we drop the subscript and denote such representations just by Γ .